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Data Collection

Although rarely discussed in research articles or book chapters, the process of data collection can pose a unique challenge to the group process researcher. Several decisions need to be made regarding the medium of data collection: Should the data be recorded or collected in real time? If recorded, are audio or video recordings necessary? Should the verbal portion of the recordings be transcribed or can coding be done directly from the tapes? Data needs and accessibility will influence the answers to these questions.

Recording versus Real time

Obtaining a recording of the group's process is always preferable to collecting all the data in real time. Ideally a researcher would observe the group interaction while recording. Recording allows one to code the data at the coder's rather than the group's pace. Observation provides an opportunity to pick up on nuances that may be lost in recording. However, at times it is not possible to record group interactions, especially when studying groups in field settings. Concerns of confidentiality often preclude recording of interaction. In many of these situations coding in real time is the only option (e.g., Komacki, Desselles, & Bowman's [1989] study of sailing crews). Using a simple coding scheme and having multiple, highly trained coders will increase the likelihood of reliable data collection. More detailed coding schemes

make real-time coding difficult as they require complex coding decisions to be made quickly. This difficulty, combined with the inability to review a past code assignment, increases the risk of unreliable coding. Use of multiple coders can reduce this concern in providing multiple assessments of the interaction.

Audio versus Video Recording

Interest in nonverbal behavior and/or in identifying speakers will drive the decision of whether video rather than audio recordings are needed. Whereas audio recordings are less expensive and easier to obtain, they preclude the collection of nonverbal behavior and make it difficult to identify speakers. However, this speaker identification problem can be overcome by using a multiple track audio-recording machine and recording each speaker on a separate track. Video recordings also allow speakers to be identified. However, camera placement is important to ensure a clear view of all group members. This might require the use of several cameras, depending on the configuration and size of the group and room. If multiple cameras are used, it's useful to link them to a common time code to facilitate integration among the separate recordings. If a video mixer is available, the separate images can be mixed onto one recording using separate windows for each image. As mentioned earlier, software developments have improved the interface between video recordings and coding equipment, making the use of video recordings more attractive.

The use of audio versus video recording is also influenced by the research setting. It is often difficult to obtain permission to videotape group processes in organizational settings. Concerns about identifiability and confidentiality are high. However, several organizations (e.g., GM, Xerox) use videotapes of group decision making as sources of feedback for managers. Tapping into ongoing efforts might increase the odds of gaining video access.

Video equipment options abound. The researcher must choose between VHS versus 8mm formats. Each format can hold two hours of recording time.⁵ Eight-mm formats (8mm, Hi-8) provide somewhat higher resolution and smaller cassette size than do VHS formats (VHS, S-VHS). While useful for original recordings, 8mm tapes are not as amenable to editing and copying, and are more likely to stretch (due to the small surface area of the tape) as compared to VHS tapes. In addition, each format has both lower- (standard VHS, 8mm) and high-

end (S-VHS, Hi-8) options. Lower-end quality is usually adequate for recording group interaction. However if images on computer screens are also to be recorded, high-end equipment is needed.

Alternative recording media are also available. For example, computer mediated communication is easily recorded using e-mail systems and software programs such as the "Electronic Conference System" developed by the University of Illinois Office for Information Management. Group Decision Support Systems software (e.g., Vision Quest and GroupSystems)⁶ also provide opportunity to record written interaction while employing the system. Handwriting recognition systems are also available for saving handwritten coding and field notes.

Transcriptions

Prior to coding recorded data, a decision must be made whether to transcribe the verbal interaction. With transcripts one can code from the transcripts alone or in combination with the recording. Without transcripts one must code from the recordings alone. Transcription, especially of videotapes, is time consuming and potentially expensive if professional transcription services are obtained. While direct coding of recordings is tempting for this reason, it is more difficult to reliably identify the units to be coded (i.e., obtain unitizing reliability) without transcripts. Consider the sample transcript in Table 5 of a group of four students working on a Tinkertoy model building task (Weingart, 1992). The interaction occurred at a very quick pace with interruptions, fragmented sentences, and short speaking turns. These speech patterns, common to groups working under time pressure, are difficult to unitize (i.e., divide into units or segments to be coded) and code without transcripts.

If one chooses not to transcribe verbal interaction, but rather work directly from the recordings, the videotapes need to be indexed using time codes. These codes are recorded onto the videotape, assigning a specific address to each video frame. These time codes can be used by the coders to identify the beginning and ending of each unit to be coded. Behavioral coding assessments can then be linked to the location on the video using the time codes. Time codes can be recorded as a counter in a box at the bottom of the picture (i.e., burned into the image) and/or as a readable code on the tape. Time code can be recorded longitudinally on the second audio track or on a dedicated time code track (LTC—longitudinal time code), or it can be recorded vertically

Table 5. Sample Transcript—Tinkertoy Building Task

Speaking Turn	Speaker	
1	2	Look at this piece—it has side openings. Should we just like make nine, nine parts?
2	4	Yeah I guess. Did you already put that on an angle?
3	3	Yeah
4	4	Okay, you did it right. I'm doing it wrong.
5	1	Do you want to just assemble these five with just the yellow rods and I'll do the cones.
6	3	Okay, I think this is the most quick though.
7	1	Yeah.
8	3	And you want me to do the yellow?
9	1	Why don't you do the five things connected with yellow?
10	3	Okay.
11	1	And I'll do everything else?
12	3	Okay.
13	1	I'm getting confused looking at this.
14	3	Really?
15	1	And it's here.
16	3	Take the glue off.
17	2	Having fun?
18	1	No, actually.
19	3	You know these things are actually a lot of fun I wish we'd had these when I was little. I don't remember having tinkertoys.
20	1	none of them?...
21	3	I was a girl. I always played with dolls.
22	1	Okay, just say...
23	3	I'm trying.
24	1	Easier put to the test. So maybe we can exchange, give me another.
25	4	Well how do you do this?

as part of the video signal (VITC—vertical interval time code). VITC must be recorded at the time the original recording is made because it is integrated into the video signal. LTC can be added to an existing videotape because it is laid on a separate track.

Coding Scheme Development

Identification of the behaviors to be studied is a pivotal step in group process research. The choice of categories in which behaviors will be

classified has a profound influence on one's ability to test and support hypotheses. Consider the coding of information exchange in negotiation. When using a coding category designed to capture information sharing of all kinds, little or no relationship between information sharing and quality of agreements was found (Weingart et al., 1990). However, after breaking the information-sharing category down further—into sharing information about one's priorities across issues (which issue is most important to me) versus information about preferences across options within an issue (which option of a particular issue I prefer)—we found the former positively influencing and the latter negatively influencing quality of agreements (Weingart, Hyder, & Prietula, 1996).

General versus Task-specific Coding Schemes

Several discussions can be found regarding how to construct a coding scheme (Folger, Hewes, & Poole, 1984; Hewes, 1979; Poole, Folger, & Hewes, 1987; Trujillo, 1986). There are two general approaches used to do so (Poole & McPhee, 1994). The first involves development of an exhaustive, logically complete, classification system that can be applied across task types (e.g., Bales, 1950; Bales & Cohen, 1979). These schemes often employ decision trees, beginning with general categorical distinctions and narrowing to more specific categorical assignments (e.g., Anderson, 1983; Futoran, Kelly, & McGrath, 1989). For example, TEMPO, a Time by Event by Member Pattern Observation system for coding interaction from videotapes, begins with two broad categories: production function and nonproduction function. Production function subcategories include (a) proposing or (b) evaluating ideas regarding the (1) content of the task or (2) process through which the task is being performed, resulting in four categories (content-propose, content-evaluate, process-propose, process-evaluate). Each of these categories has three or four subcategories. For example, process-propose subcategories differentiate between comments on (a) goals, (b) strategies, and (c) acts designed to keep the task moving. Each subcategory can be broken down further to capture performance of a particular task. The goal of such a coding scheme is to capture on-task and off-task behavior using a time-based system that is applicable to different kinds of groups and tasks.

The second approach indexes behaviors associated with performing the specific task at hand. For example, a coding scheme of groups

performing a structure-building task included quality talk, planning, monitoring, discussion of acts and supplies, extra-role interaction, morale-building communication, negative communication, nontask-related, and uncodable comments (Weldon, Jehn, & Pradhan, 1991). While several of these categories can generalize to other types of group tasks, the majority of the categories were specifically derived from theory about groups performing production tasks (Weldon & Weingart, 1993).

Although the generalizability of results using a task-specific approach is more limited than using a generic coding scheme, it complements the use of more domain-specific theories of group behavior called for by Hackman and Morris (1975) and Goodman, Ravlin, and Schminke (1987). In a review of several models of group effectiveness, Goodman, Ravlin, and Schminke (1987) point out that although process plays a major role, the construct is expressed at a level of generality that makes it very difficult to study. Due to the large number of factors that might influence the interdependencies among group members (e.g., task, technology, organizational context), it seems difficult, if not impossible, to generalize findings about the group's process from one type of task to another. Hackman and Morris (1975) also recognized this limitation of general models of group effectiveness, and in response, suggested the need for "smaller theories" relevant to performance effectiveness under specified circumstances.

Theoretically Derived versus Data-derived Coding Schemes

Task-specific coding schemes can be either theoretically derived (e.g., Weingart, Hyder, & Prietula, 1996) or developed using a more grounded approach, derived from observation of task performance (e.g., Hawes, 1972; Hirokawa & Pace, 1983). However, the distinction between using traditional theory-driven versus a more data-driven approach is necessarily blurred in practice. Human behavior is typically too complex to anticipate all relevant behaviors without some direct experience with the group interacting on the specific task at hand. While theoretically a coding scheme may make sense for groups in general, how those behaviors are actually enacted in the group will depend on the task at hand. Thus, a more ideographic approach, recognizing that the meaning of words or actions might only be known by members of the group or those experienced with the task, is sometimes taken.

Benefits of both theory and application can be obtained by using a hybrid strategy of coding scheme development. The first step in this strategy is to develop categories based on theoretical predictions of the types of behavior that are expected to be important. Several questions should be considered at this point. (1) Should the categorization scheme be exhaustive? Both theoretical and analytical concerns are relevant to answering this question. If interested in detailed interaction patterns, coding all verbal behaviors may be important. However, if some behaviors are not theoretically interesting, but all behavior must be classified (such as when conducting a sequential analysis), the use of a "miscellaneous" or "other" category can be used. (2) How detailed should the scheme be? The answer to this question depends on the goal of the analysis—whether global or detailed interaction patterns are of interest. Methodological concerns are also relevant. As the number of categories increases, the risk of combining two behaviors that potentially serve different purposes is reduced. However, the downside of the proliferation of categories are twofold. First, coding becomes more difficult as categories become less distinct. Coding errors are more likely, lowering reliability, and potentially necessitating the collapsing together of categories. Second, as the number of categories increases, the frequency of behavior in each category necessarily decreases. Low frequency within categories is problematic for most statistical methods—especially sequential data analysis techniques.

When designing a coding scheme the researcher should develop a coding manual, that is, a set of application rules/guidelines for the coders, including identification of the unit to be coded (thought, sentence, speaking turn, paragraph), category labels, definitions, and rules of thumb for distinguishing between categories and using context (i.e., statements surrounding the unit of interest) to interpret meaning. Determining whether to allow the use of context in coding the meaning of a statement is a fundamental decision. Certainly examining the context is useful, and may be necessary, in ascertaining the meaning of statements (McGrath, 1991). For example, examination of a previous statement can inform the coder whether the statement "yeah" denotes agreement with the previous statement versus understanding of what has been said. Speaking turns 3 and 7 in Table 5 are examples of how context can aid interpretation. The "yeah" in turn 3 is a clear agreement with the statement "Did you already put on that angle." The "yeah" in turn 7, in response to "Okay, I think this is the most quick though," is not so clear, potentially denoting agreement *or* understanding. Even

when faced with such ambiguity, coders often feel the need to categorize the statement. Decision rules for ambiguous statements need to be developed to avoid over- or misinterpretation of the data.

After developing a theoretically based coding scheme, the second step of a hybrid strategy involves refining the coding scheme through an iterative process. Rarely, if ever, can a novel coding scheme be employed without fine-tuning. Application of the scheme will help in revealing categories that are too broad (when behaviors that seem quite different are similarly classified), and others that are too narrow or too similar to other categories (when deciding between two categories for a given behavior is repeatedly a problem). Relevant questions include: Can coders differentiate between the categories when applying them? Are there categories for all relevant behaviors? Can the scheme be applied reliably (discussed below)? If not, is it because of problems in training the coders or in the design of the scheme?

Coding Scheme Development and Coder Training

Most questions regarding the coherence of the coding scheme can be addressed during coder training. Coders, naive to the theory and hypotheses, will not hold the same understanding of the process as the researcher, resulting in the need for clarification and elaboration of coding information. Coders should be considered part of the research-design effort, helping to resolve problems in applying the coding scheme. Thus, the coding manual should be considered a living document, open to revision and clarification during the development and training process, but then locked in when actual coding begins.

During training, problems involving coding scheme application versus design can be distinguished and resolved. Sample coding can be compared between the coders and the developer of the coding scheme. If disagreements are systematic and easily resolved, then the problem probably resides in the coders' application of the coding scheme and can be fixed through clarification of definitions and additional training. However, if disagreements appear to be haphazard and difficult to resolve, then the problem may reside in the design of the coding scheme itself. Categories might overlap or be missing.

One helpful technique to identify where overlap or missing categories might exist is to develop a two-way matrix with the coding categories (1 to n) on each axis, the x-axis representing the coding of Coder A and the y-axis representing the coding of Coder B. After the two coders

have independently coded a sample interaction, a hashmark representing the assignment of each unit by each of the coders is entered into the appropriate cell of the matrix. If the two coders are in complete agreement, all entries will be on the diagonal of the matrix. Disagreements will fall on the off diagonal. Off-diagonal cells with many hashmarks signal which categories are repeatedly being confused with one another, and what the patterns of confusion tend to be.⁷ In some instances, highly confused categories need to be combined, in others, definitions need to be refined. Concerns for precision of codes, driven by the theory, should guide the decision of which route to follow.

Coding Data

There are two discrete steps in coding process data: (1) unitizing the data, or identifying the unit of analysis to be coded, and (2) content coding the data, or applying labels to units as a means of capturing what occurred. Next, I discuss issues of unitizing and unitizing reliability, followed by a discussion of content coding reliability and validity.

Unitizing

The unit of analysis in interaction analysis can vary from a single utterance that contains meaning to an interact across two individuals. Speaking turns, units that include all actions and/or statements made by an individual while he or she holds the floor, are often used when studying interpersonal interaction because they are predicated on identification of and changes in speakers, which are central to many theories of group decision making and performance. Acts focusing on an identifiable idea or thought have been used when interested in the content of what's being said rather than the interactive nature of the group process. Decisions regarding choice of unit type must be linked to the research question being asked, with special sensitivity to the appropriate level of analysis and where in action and speech relevant meaning resides. By selecting too small of a unit, meaning can be lost (by parsing statements that together convey a different meaning than when separated) or redundancy added without meaning (by separating immediate restatements that are completely repetitive rather than clarifying).

Consider the following speaking turn from a negotiation, "I propose we order the parts from the Wexford supplier; the other supplier is too

expensive. I also want a 10 percent volume discount." If the unit of analysis is the act, three units would be identified, each receiving a separate code.

1. I propose we order the parts from the Wexford supplier (*one-issue offer*);
2. the other supplier is too expensive (*substantiation*).
3. I also want a 10 percent volume discount. (*one-issue offer*).

However, if the unit of analysis is the speaking turn, this group of statements would be coded as a two-issue offer (statements 1 and 3) and a substantiation (statement 2). In this case, the use of the act as the unit of analysis would decrease the identification of multi-issue offers and increase single-issue offers, thus misrepresenting the data. This difference in coding is notable because negotiation research suggests that multiple-issue offers are more likely to increase joint gain than single-issue offers (Weingart, Hyder, & Prietula, 1996).

A related issue that can complicate unitizing is the occurrence of backchanneling and interruptions. Backchanneling often appears as nonsubstantive interruptions, which communicate that the receiver understands what is being said, but does not imply agreement (e.g., "uh-huh," "okay," or "I see"). If identified as a substantive unit, backchanneling might parse meaningful statements into uncodable fragments. Thus, a decision must be made whether to code these utterances as identifiable units. While nonsubstantive interruptions can often be disregarded, substantive interruptions usually cannot. Because substantive interruptions potentially disrupt the flow of communication from another group member, they should be treated as identifiable units.

In contrast, when units are larger and only one code is allowed per unit, some data can be lost. If multiple categories of statements are made during a speaking turn, the researcher must decide which code best represents the behavior within a given unit. Dominance schemes, identifying which behaviors are expected to have the greatest impact on the interaction, can be developed to provide rules of thumb for these decisions (e.g., Weingart, Bennett, & Brett, 1993). Alternatively, the first or the last code within the unit might be retained. Regardless of the approach that is used, the risk of losing valuable information remains.

Consider the first speaking turn in Table 5. Two distinct actions occurred during that speaking turn (using Weldon, Jehn, & Pradhan's [1991] coding scheme: discussion of supplies and planning). When

conducting a sequential analysis at the speaking-turn level of analysis, one of these actions needs to be dropped from the data set. A dominance scheme might suggest that planning is dominant over discussion of supplies. However, if first statements within a speaking turn were retained, that unit would be recorded as a discussion of supplies.

Unitizing Reliability

Unitizing reliability, the degree of agreement regarding identification of the units to be categorized, is most typically assessed using Guetzkow's *U* (Folger, Hewes, & Poole, 1984; Guetzkow, 1950). Guetzkow's *U*, a measure of disagreement between two coders, calculates the difference between the number of units identified by an independent coder and the "true" number of units (the average of two coders' estimates), using a simple equation

$$U = (O_1 - O_2) / (O_1 + O_2)$$

where $U \times 100$ is the percent of disagreement and O is the observer or coder.⁸ One shortcoming of this measure is that it does not identify unit-by-unit agreement, but rather focuses on agreement regarding the total number of units. Folger, Hewes, and Poole (1984) suggest the need for use of a unit-by-unit agreement measure when the units are subjective, the coding scheme is not exhaustive, or the data are to be used in sequential analyses.

Content Coding and Interpretive Reliability

Issues of developing a content coding scheme were discussed previously. In this section and the next, issues of application of the coding scheme, or interpretive reliability and validity, are discussed.

Interpretive reliability refers to consistency in applying labels to the units. Several factors contribute to the ease of obtaining interpretive reliability. In addition to identifying categories to be applied, coding schemes should provide decision rules for interpreting actions and dialog to aid in the application of those categories (Folger, Hewes, & Poole, 1984). While a cleanly unitized transcript will remove unreliability of units from assessments of reliability of content coding, improvements in definitions of categories and the rules for their application will increase the odds that two (or more) coders will similarly categorize given behaviors.

Interpretive reliability is typically measured across coders. That is, multiple coders independently code the same text, and their assignment of categories are compared for consistency. Consistency measures are also usefully obtained within coders across time to check for drift in application of the coding scheme. Both global and category-by-category measures have been used. The most common global measure is Cohen's kappa (1960)

$$\text{Kappa} = (P - P_c) / (1 - P_c)$$

where P is the observed percentage agreement among coders and P_c is the proportion of chance agreement. The benefit of using kappa over straight measures of percentage agreement between coders is its correction for agreements due to chance. Cohen's kappa has been found to be one of the most complete measures of interrater agreement when compared with a host of published indexes across a wide range of criteria (Popping, 1988). However, one disadvantage of this and other similar (i.e., non-weighted) measures is that lower than would be expected estimates of reliability result when coding categories are not used with the same frequency. This occurs because more agreement is attributed to chance for the dominant categories than is necessary. When it is expected that observations will tend to be greatly skewed across categories with a given coding system, it may be advisable to use a weighted kappa (Cohen, 1968).⁹ Extensions of Cohen's kappa for more than two coders, estimates of variance, and statistical tests can be found in Fleiss (1971).

Cohen's kappa can also be applied on a category-by-category basis. Rather than assuming that the coding in each category is equally reliable, category-by-category measures assess each category independently. Calculating category-by-category kappas aid the researcher in identifying unreliable categories that could undermine data analysis, but that otherwise would be missed if pooled with other more robust categories (see Folger, Hewes, & Poole, 1984 for an interesting discussion).¹⁰

Interdependence of Unitizing and Interpretive Reliability

As discussed earlier, high reliability is important in both steps of the coding process: unitizing and content coding. The reason for seeking high unitizing reliability is to ensure that multiple coders of the same

interaction will be coding from the same set of units. To ensure high interpretive reliability it is best to conduct unitizing and coding in separate passes because errors in unitizing will have a strong impact on interpretive reliability. This is especially important when coding from recordings where it is more difficult to identify the specific unit to which a code has been assigned.

Consider the following situation. Two coders simultaneously unitize (by speaking turn) and code a videotaped interaction. They record on paper the speaker, time the speaking turn began, and a code representing the content of the statements. When it is time to check for interrater reliability, it is discovered that the speaking turns were relatively short, the conversation quickly jumped from one speaker to the next, interruptions and backchanneling were common (see Table 5 for an example), and as a result the two coders have identified the speaking turns slightly differently. Because the units (speaking turns) are now difficult to match across the coders, it is next to impossible to check for interrater interpretive reliability. This problem could have been avoided by completing the unitizing first, checking for unitizing reliability, and then content coding from the same set of pre-unitized tapes or transcripts. Unfortunately, there is little to do in such a situation but discard the existing coding and begin again using a two-phase approach—unitize first, code second.¹¹

Interpretive Validity

Concerns of interpretive validity relate to the degree to which a coding scheme taps into the information it was designed to obtain. Different approaches have been used to validate coding schemes, depending on the goals of the research. Poole and McPhee (1994) identify three types of claims that might be made in an attempt to demonstrate validity of a coding scheme. These claims are linked to the source of interpretation of the actions being studied: (1) observer-privileged: where the researcher seeks to explain interaction from the outside without reference to subjects' perspectives, (2) generalized subject-privileged: focusing on shared meanings for members of a culture, and (3) restricted subject-privileged: with the goal of identifying idiosyncratic meanings for people in a particular group or relationship. As one moves from observer to restricted subject-privileged approaches, more input from participants and more intimate knowledge about the work group (as might be experienced by a group member) is required.

While most group process research using interaction analysis uses an observer-privileged mode (e.g., Kiesler & Sproull, 1992; Pruitt & Lewis, 1975; Weingart, 1992; Weingart, Bennett, & Brett, 1993), they typically aspire to obtaining generalized subject-privileged validity in their attempts to tap the shared meaning of group member behavior. For example, in coding tactical behavior in negotiations, offers, different types of information exchange, substantiation of positions, threats, and so forth are typically coded. These coding schemes are derived from theory and research about negotiation behavior, suggesting that the researcher (observer) has an understanding about tactical behavior that is meaningful, independent of the intent and perceptions of the negotiators themselves (subject-privileged). However, what might be coded as "information sharing about a negotiator's intent" might actually be perceived by the other party as a "threat," resulting in an invalid categorization. Unfortunately, this type of validity is rarely assessed.

Data Analysis

Several different techniques are available for the analysis of categorical data that either aggregate the data into frequency counts or assess sequential patterns. Regression analysis has been used to assess the effects of absolute and relative frequencies of behavior on outcomes of interest. Markov chains, lag-sequential analysis, phasic analysis, log-linear models, and optimal matching analysis have been used to identify sequential patterns in the data.

Frequency Analysis

Frequency analysis uses absolute counts or relative frequencies to capture the composition of relevant communication and/or behavior. Absolute frequencies provide information about the frequency of occurrence of a given behavior, independent of the frequency of other behaviors that might have occurred. Relative frequencies, which control for the total amount of behavior, provide information about the proportional use of a given behavior. Use of relative frequencies controls for verbosity of the group and assumes that a speech act that occurs within a short interaction will have a stronger impact than the same speech act that occurs within a longer interaction.

Consider two performance groups that differ in their level of verbosity. The more verbose group talks for a longer period of time

than the less verbose group, engages in more total planning, but also engages in other behaviors with greater frequency as well. When using an absolute measure of behavior the data would suggest that the more verbose group engaged in more planning. In contrast, when using a relative measure both groups might appear to have engaged in (roughly) the same amount of planning. Which measure is a more accurate indicator of planning *per se*? Again, one's theory of group behavior must guide the answer to this question. For example, if it is believed that a given behavior can influence a group outcome regardless of the amount of other behavior present, absolute frequencies are more reflective. However, if one is interested in the relative impact of a behavior within the overall interaction, relative frequencies might be more appropriate.

Typically, frequencies of behavior are related to an independent and/or dependent variable using regression techniques. Questions in the performance group setting that have been asked using this approach include: How do group goals influence the amount and type of planning in groups. How does the amount of planning affect quality and quantity of group performance? How does the amount of social communication relate to productivity and or cohesiveness?

Sequential Analysis Techniques

Sequential analysis is predicated on the belief that social interaction has a temporal form. Rather than aggregating data across time, relevant behaviors are treated as discrete units, with the goal of finding recurring patterns in the data. Patterns might result from specified inputs to the group, or predict performance outcomes. Popular analytic techniques include lag sequential, Markov chain, phasic analysis, and log-linear analysis. Generally, each of these techniques looks at transitions from one state (e.g., type of behavior) to another. Recurring patterns are then identified and tested for significance. Below I discuss each technique briefly. For more detailed information, Hewes (1979) provides a friendly introduction and Gottman and Roy (1990) provide a useful guide to sequential analysis for the behavioral sciences.

Lag-sequential analysis (LSA; Sackett, 1979) identifies systematic patterns of behaviors by comparing an observed pattern of behavior to the possibility of it occurring by chance. The unit of analysis is the transition from one behavior to another. A lag is the number of steps (behaviors) that separate a particular behavior from a criterion event.

Although reformulated to correct for bias in the sampling method (Allison & Liker, 1982; Morley, 1987), criticisms remain (Kellerman, 1988). These criticisms are largely based on the fact that lagged behavior (i.e., the probability that a certain behavior at, say, time $t + 4$ follows a given behavior a time t , without considering what occurred in the interim), rather than full sequences, are examined. For this reason, when used by itself LSA provides "(1) an incomplete description of the pattern of dependencies in sequentially ordered categorical data, and (2) a potential for inaccurate description of the pattern of dependencies" (Kellerman, 1988, p. 2). While useful in investigating prototypical sequences of acts, Kellerman (1988) suggests LSA be used in conjunction with Markov analysis to compensate for its weaknesses.

Markov models assume that, at any point in time, the probability of the next behavior occurring depends on behaviors that occurred previously. This number of previous behaviors required to predict the current behavior is called the *order* of the chain (see Gottman & Roy, 1990). For example, in a first-order Markov chain the distribution of the current behavior depends only on the immediately previous behavior. Current behavior in a second-order chain depends on the two immediately preceding behaviors. The behavior that occurs at a given point in time is often called the *state* of the process at that time. Order then is the number of past states affecting the current state. Transition matrixes comprised of probabilities that current states follow prior state(s) are analyzed to determine the sequential structure in the data. Through the use of resampling methods (Efron & Tibshirani, 1993), standard errors can be calculated such that patterns of behaviors can be statistically compared across experimental conditions. For example, Weingart, Prietula, and their colleagues (1996) used Markov models to test differences in tactical patterns of negotiators with and without tactical knowledge.

Markov analysis is useful in that it taps nicely into the responsive nature of interaction (Poole & McPhee, 1994). However, one assumption underlying the use of Markov models, that communication can be modeled as an orderly series of actions and reactions, has been questioned by communication researchers (Hewes, 1979; O'Keefe, Delia, & O'Keefe, 1980). In addition, studies that fit Markov models to data without considering theoretical assumptions about the interaction have been criticized (see McPhee & Poole, 1981). More recent research using Markov chains have attempted to respond to these concerns (e.g., Capella, 1979; Hewes, 1980; Weingart, Prietula et al., 1996).

Phasic analysis provides an alternative to Markov chain analysis (Holmes & Poole, 1991). In phasic analysis social events are believed to be divided into phases which may be linear or cyclical in nature. Process dynamics drive the shifts in behavior from phase to phase. These phases, or episodes, consist of unified sets of acts that serve a particular function. This approach is based on the assumption that behavior can be meaningfully described in units larger than individual acts. In addition, phase mapping can be flexible, that is, phases can vary in length. For these reasons, phase analysis is not as directly tied to apparent behavior patterns as are Markov models, and therefore free from the criticisms that Markov analysis faces.

Log-linear analysis, a generalization of analysis of variance to multi-way contingency tables, is a method that can be used to determine the impact of experimental factors on specific sequential patterns (Gottman & Roy, 1990). Hierarchical models are most commonly used (ones that involve all main effects and interactions), and though somewhat difficult to interpret, are superior to using regression analysis of dummy variables when analyzing binary data (Knoke, 1975).

One final analytic technique, optimal matching analysis, determines similarities between sequences drawn from relatively small sets of possible elements (e.g., behaviors) (Sankoff & Kruskal, 1983). The similarity between any two sequences can be conceptualized as a "distance" between the sequences. The distance between any two sequences is computed using the number of insertions, deletions, and substitutions required to transform one sequence into the other (Poole & Holmes, 1995). Optimal matching analysis uses dynamic programming to compute the most efficient transformation possible. While more commonly used in genetics and computer science, Poole and Holmes (1995) have applied optimal matching analysis to comparisons of decision-making processes across groups with and without GDSS.

CONCLUSION

The goals of this paper were twofold—to demonstrate the contribution of studying dynamic group processes and to provide guidance regarding how to do so. Although recognition of the importance of understanding group interaction patterns in the field of organizational behavior is not new (e.g., Hackman & Morris, 1975), little extant research has gone

beyond conceptualizing group process as an aggregate of behavior. Group process has more typically been assumed based on group outputs, or inferred from more distal measures such as group member perceptions of the process. While organizational researchers are beginning to show renewed interest in opening the "black box" of group processes, most hesitate because of the belief that the work is too time consuming or that rigorous methods are not available.

Many of the methodological issues faced by group process researchers are similar to those faced when using other methodologies (e.g., reliability and validity of measurement). However, organizational researchers are largely unaware of the specific techniques available for studying (static or dynamic) group processes. Several factors contribute to this lack of knowledge. First, graduate training programs in organizational behavior rarely provide training in these methods. Second, little has been published in the organizational literature regarding how to do the research. One must peruse the communication literature to find relevant methodological pieces. Third, and most importantly, to date little organizational behavior research on groups has identified this approach in examining dynamic processes in groups.

Although there is a lack of awareness, it is not for a paucity of available methods and tools. Several resources provide information regarding the ways and means of studying group process. Guidelines for collecting, coding, and analyzing group process data can be found. However, because many are written for a different audience (communication rather than organizational researchers), their use requires some translation and interpretation. Although the methods are time consuming and somewhat complex, the potential payoff is high. The data obtained is rich, open to analysis and interpretation at multiple levels. Analytical tools are available to do so.

To demonstrate the added benefits of using this conceptual and analytical approach to the study of groups, consider advances in negotiation research, where these methods have been applied. Early (and continuing) negotiation research using an input-output model provided the field with knowledge regarding factors that improve the quality of agreements for the group as a whole (e.g., high aspirations, trust, problem-solving orientations). Processual research in this domain has gone further to determine the behaviors through which these effects occur (e.g., sharing of information about priorities, lowered amounts of substantiation, multi-issue offers, simultaneous issue consideration) and the processes through which these behaviors are enacted

(reciprocity, second-order Markov chains, attack-defend versus attack-attack cycles, phases of conflict resolution). Using data obtained from videotaped and live interactions, sometimes transcribed and always coded, Markov-chain, lag-sequential, regression, and log-linear analysis techniques have been used to gain better understanding of these processes. Results from this research are moving us toward greater insight, descriptive conclusions, and prescriptive advice. Next steps include a focus on convergence of tactical behavior across negotiators over time and phasic analysis of multiparty negotiation.

I challenge group researchers to consider the potential of examining dynamic processes in light of advancements in technology, methodological techniques, and analytical tools. Dynamic processes are relevant to a broad range of research topics. Consider the impact of diversity in group composition (along various dimensions) on the performance of the group, including quality of agreements and products and level of creativity. A dynamic approach could address how the application of skills and knowledge (and other cognitive factors) mediate this relationship. One might consider diversity influencing (1) the range of information accessible from the group as a whole (absorptive capacity: Dahlin & Weingart, 1996), (2) how the group identifies what information is available and who holds critical information (transactive memory: Liang, Moreland & Argote, 1995; Moreland, Argote, & Krishnan, 1996; Wegner, 1986), and how group members develop a shared representation of the problem (collective cognitive maps: Langfield-Smith, 1992). Each of these social cognition constructs have largely been studied as summary variables—that is, as static in nature. As a result, many interesting questions remain unanswered: What is the process through which group member knowledge and skills are accessed by the group? How do group members learn where knowledge resides within the group? When and how often does new information need to be presented before it is embedded in the group's knowledge base? How do individual group members' cognitive maps converge to form a collective cognitive map? Direct examination of interaction between group members should help us answer these questions.

Another potential group process issue involves how interdependencies, inherent in the design of the task, impact the development of plans for action. A dynamic process research agenda can address questions such as: How are group strategies developed? That is, how and when should planning occur? How does the nature of the group's approach to

performing the task influence the way in which plans are developed? Is there an optimal timing and sequence of information identification and exchange during the planning process? Does group consensus on a strategy need to occur for it to be effectively implemented? If so, when? Research on planning in groups has begun to address some of these questions (Weingart, 1992; Weldon, Jehn, & Pradhan, 1991). However, these studies have not used a decidedly dynamic approach to these questions.

We can look more deeply into the relationship between group norms and the level and coordination of group member effort. The task performance literature has a tradition of looking at the exertion of effort in groups, specifically focusing on the coordination of effort as an important influence on group performance. From a more dynamic perspective, one might ask: How do group members adjust to one another when trying to coordinate their efforts? What role do norms play in determining the attention group members pay to one another in determining appropriate effort levels over time?

This is an exciting time for groups researchers in organizational behavior. There is much to be learned, and many options for attacking these thorny issues. As more of us are willing to take on the challenge, and encourage our students to do the same, the social factors that have in the past served to limit this research can be reversed. Advances will require organizational support, cooperation, and access to extant work groups. The current move toward group-based work in organizations should provide impetus for needed cooperation. Investments in university observation labs and equipment for field observation are also required. These one-time investments can provide facilities for several researchers to share. Together we can figure out how groups get their work done—the question “How did they do that” will be a tractable one.

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NOTES

1. This is in contrast to a mediator which channels variance from an input variable to some outcome. Mediators are modeled as partialled effects in regression analysis